

Benefits of Using the Latest Low-Power Microcontrollers

How do the microcontrollers that have been developed specifically for low-power applications benefit designers?



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Designers are looking for low-power microcontrollers for a number of reasons. The primary reason is to extend battery life. Applications such as wireless sensors consume a lot of power while processing and transmitting the data they collect. The value of the sensor increases the longer it operates, so extending the battery life becomes important.

Another reason is that designers want to do more with their microcontrollers, while staying within their energy budgets. In the past, when dealing with a tight budget, the microcontroller would have to sleep more often. Advances in microcontrollers have dramatically dropped their RUN currents, allowing the microcontroller to RUN more often, thereby allowing more features to be designed into the application's software.



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The low power in a microcontroller gives designers an edge to work on various applications. They get a wide range of advantages like low-power libraries, gate-level optimisation and power gating. Performance-per-watt levels have evolved in many embedded applications to enable power-efficient designs that help the end user to conserve energy. In fact, considering the looming power crisis, regulatory authorities have made it mandatory for consumer appliance companies to upgrade their products and display the energy efficiency ratings, enabling the consumer to make smart choices. Energy efficiency is the need of the hour and low-power microcontrollers are the foundation for efficient devices and appliances.



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Energy efficiency and power management are the key technology trends that designers are looking for. Across all electronic devices—from consumer electronics and household appliances to manufacturing equipment and metering—low-power microcontrollers help in increasing power efficiency by harvesting and effectively using power, thus making the systems more cost-effective. Low-power microcontrollers also help designers to develop environment-friendly products, as low power helps in reducing the effect of radiated emissions.



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Consumers and industrial companies, alike, are increasingly looking towards wireless and battery-backed instruments. It is therefore important for each component in a battery-operated system to consume as little power as possible, as this will translate into a smaller power source and indirectly lower component cost and PCB size. From the microcontroller design point of view, recent developments in advanced low-power microcontroller technology have enabled designers to attain high microcontroller processing performance as well as low-power consumption in standby modes. This helps to maximise the battery life in these portable battery-operated and battery-backed applications.

Since a microcontroller can spend substantial time being inactive, many designers use a 'power budget' to determine the average power consumption and to calculate the battery requirements. What are your thoughts on this?

Microchip's free Battery Life Estimator provides microcontroller models to generate a report of the average current, the life of a battery in that configuration, and whether the battery will support the modelled application. This allows designers to try different approaches to save power, whether it is better to run a portion of their code at full speed to finish faster, or if running slower over a longer time gives the lowest power consumption. (Note that this is only a model. There are a lot of other variables involved, so bench trials are always important.)

Yes, designers are using the power budget these days to determine the power consumption and also to improve and calculate the battery's performance and its further requirements. This is a good practice as one can check the performance of the battery and know exactly where it lags (if it does) behind. This ensures a good quality of battery and also a longer life cycle.

The power budget calculation is never an easy exercise. There are many aspects that need to be considered while working out the battery-life estimate. Important parameters include (i) battery self-leakage, (ii) duty-cycle calculation (in active and sleep modes), (iii) time duration taken to switch between different modes, (iv) optimised voltage and frequency levels, (v) peripheral circuitry-power calculations (Iq currents of active devices).

Power budgeting is certainly one of the methods used to estimate the average power consumption. But one should not take this as accurate and make critical recommendations about a system's operating hours based on a particular battery type. Such estimation should take into consideration other components on-board too, and designers should not make calculations based only on aspects of the microcontroller.